MTP: Multi-Token Prediction Layer

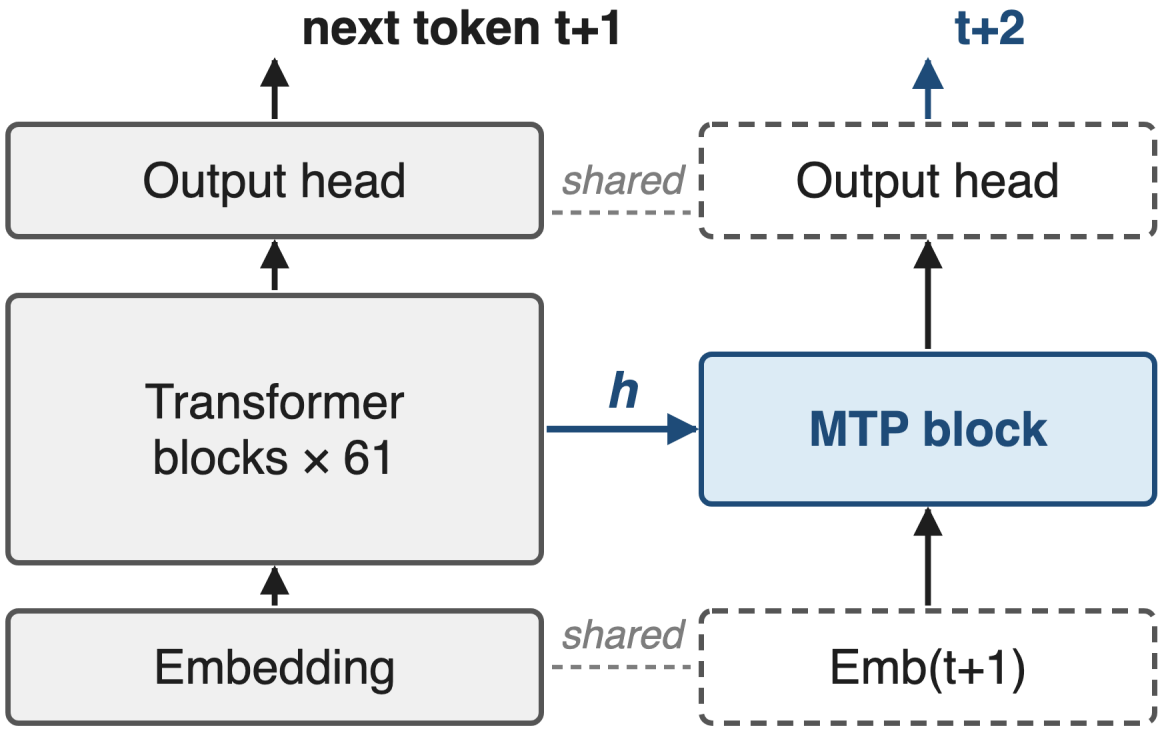
One extra block: denser training signal, built-in drafter

DeepSeek-AI (DeepSeek-V3/V4) · GLM-4.5 / GLM-5 · Megatron-Core

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Frontier MoEs ship one extra MTP layer

- DeepSeek-V3/V4, GLM-4.5/5, Kimi K3: **1** MTP layer each
- DeepSeek-V3: **14B** of its **685B** params
- Megatron-Core: MTP + MLA + CP together



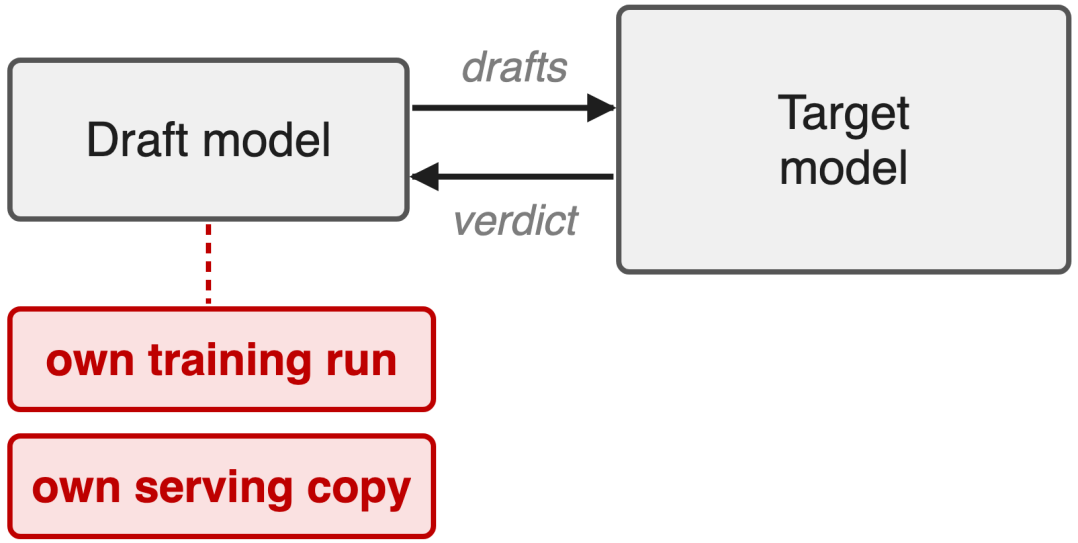
Decoding still emits one token per pass

1 token

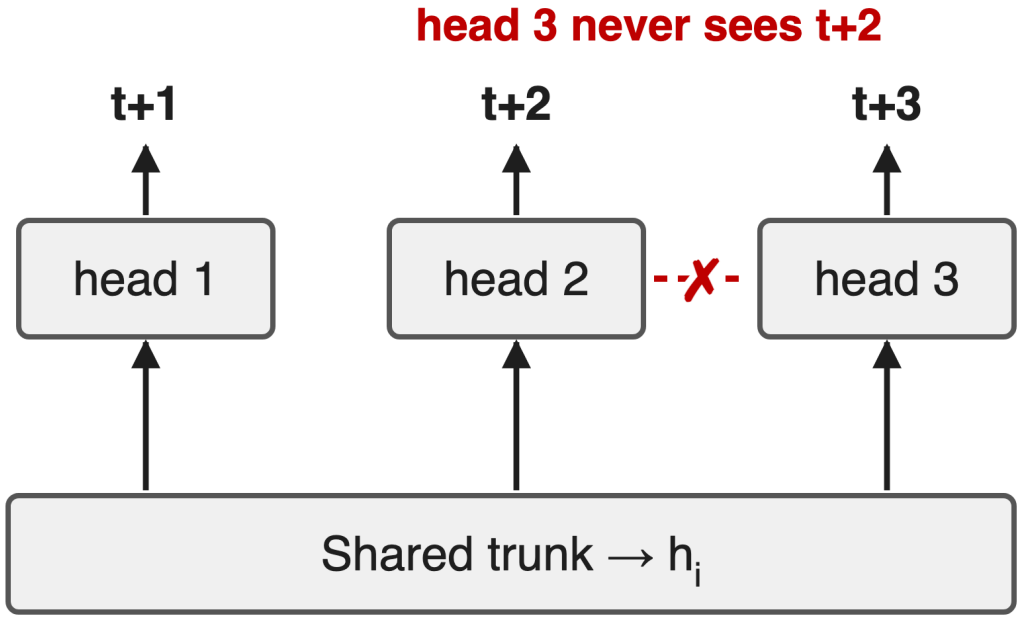
per forward pass through all 61 layers of DeepSeek-V3 (37B active), and next-token training gives each position one target

DeepSeek-V3, arXiv 2412.19437 §4.2; autoregressive decoding: arXiv 2211.17192 §1

Bolt-on drafters and parallel heads fall short



(a) Separate draft model



(b) Parallel independent heads

One adds a second model; the other breaks the causal chain

Insight: train the drafter inside the model

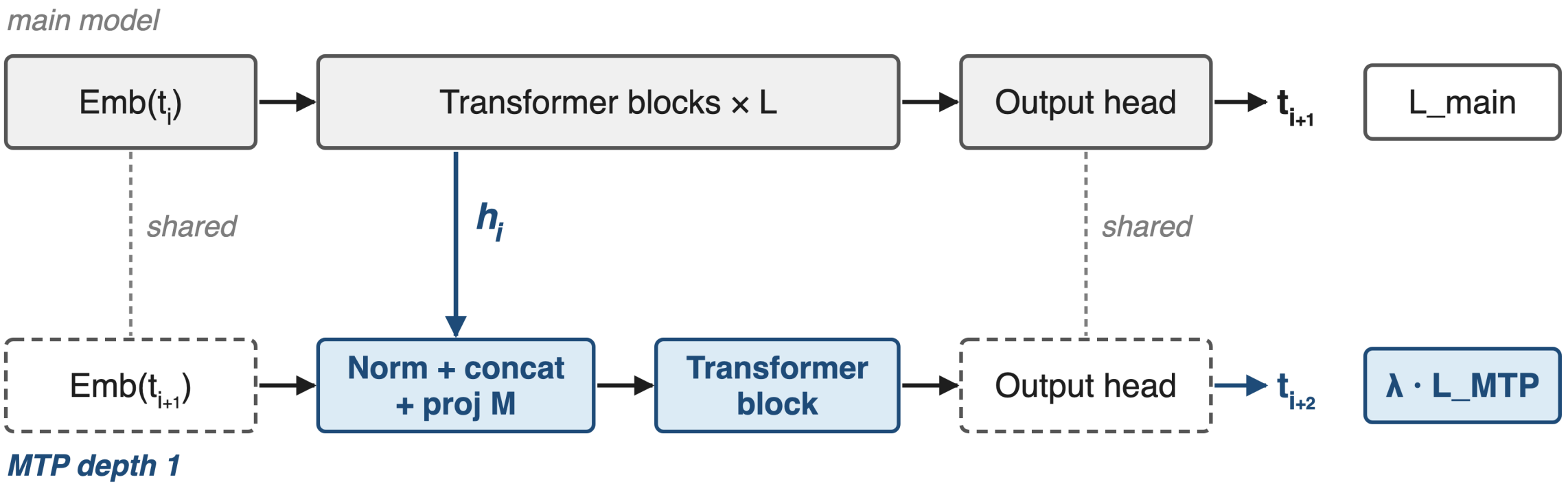
- One extra block reads **h** and the next token
- It learns the token after next, during pretraining

Insight: train the drafter inside the model

- One extra block reads **h** and the next token
- It learns the token after next, during pretraining

MTP drafts are accepted 85–90% of the time: one block, two jobs

One MTP depth reuses embedding and head



Loss = L_main + (λ / D) · Σ L_MTP,k (DeepSeek-V3 Eq. 25); dashed boxes reuse the main model's weights

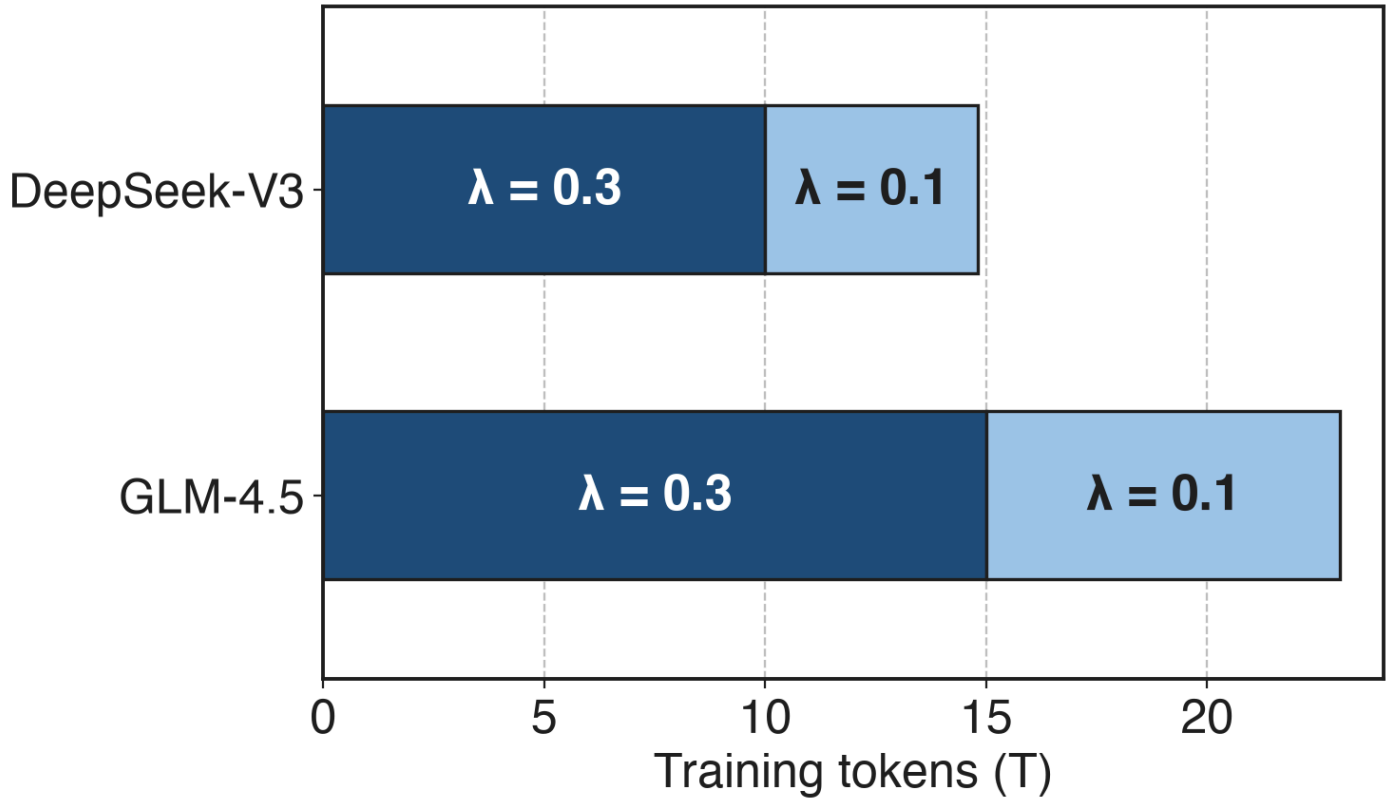
❶ Each depth chains h to the next token

	i = 1	*i = 2*	*i = 3*	*i = 4*
token t_i	The	cat	sat	on
main target t_{i+1}	cat	sat	on	the
MTP input	h(The) + cat	h(cat) + sat	h(sat) + on	h(on) + the
MTP target t_{i+2}	**sat**	**on**	**the**	**mat**

At i = 2, MTP gets h(cat) plus "sat" and must predict "on"

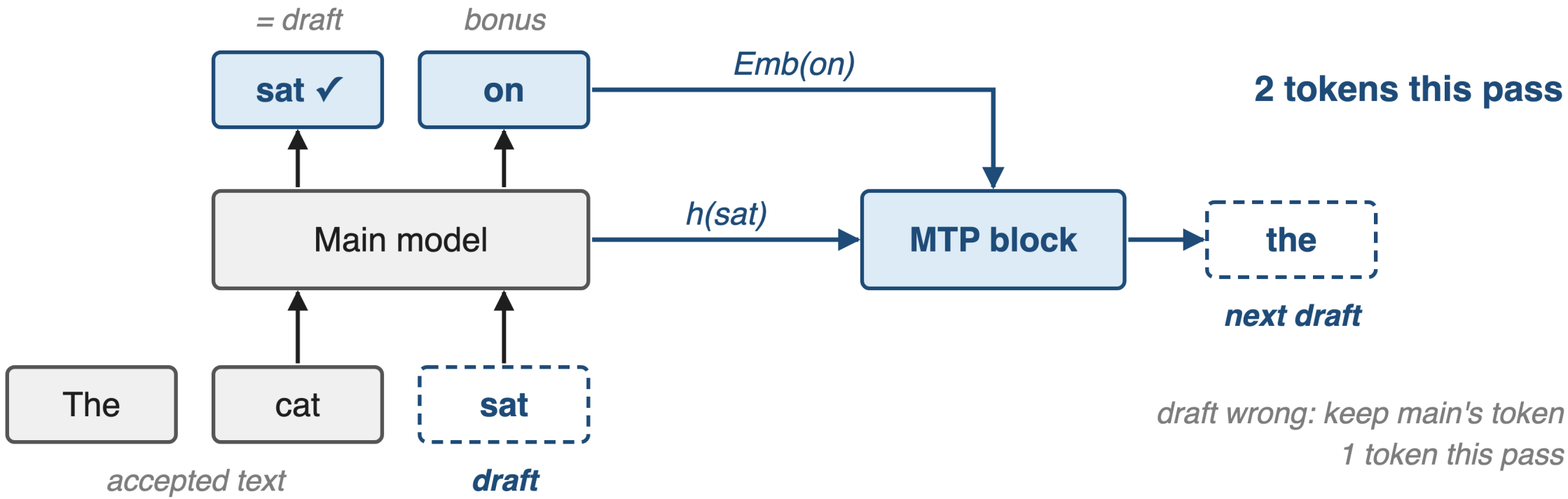
❷ λ weights the MTP loss: 0.3, then 0.1

- Total: L_main + λ · L_MTP
- DeepSeek-V4: **0.3**, then **0.1** at LR decay
- Megatron-Core default: **0.1**



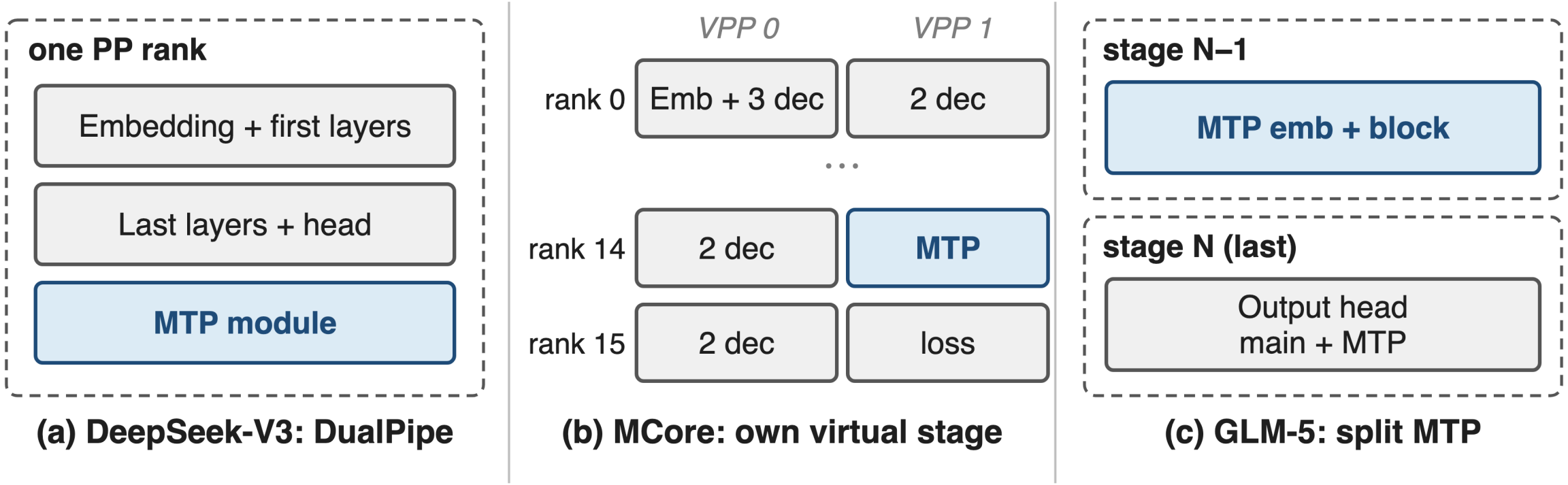
Production settings, not a tuning result

❸ At decode, MTP drafts and the model verifies



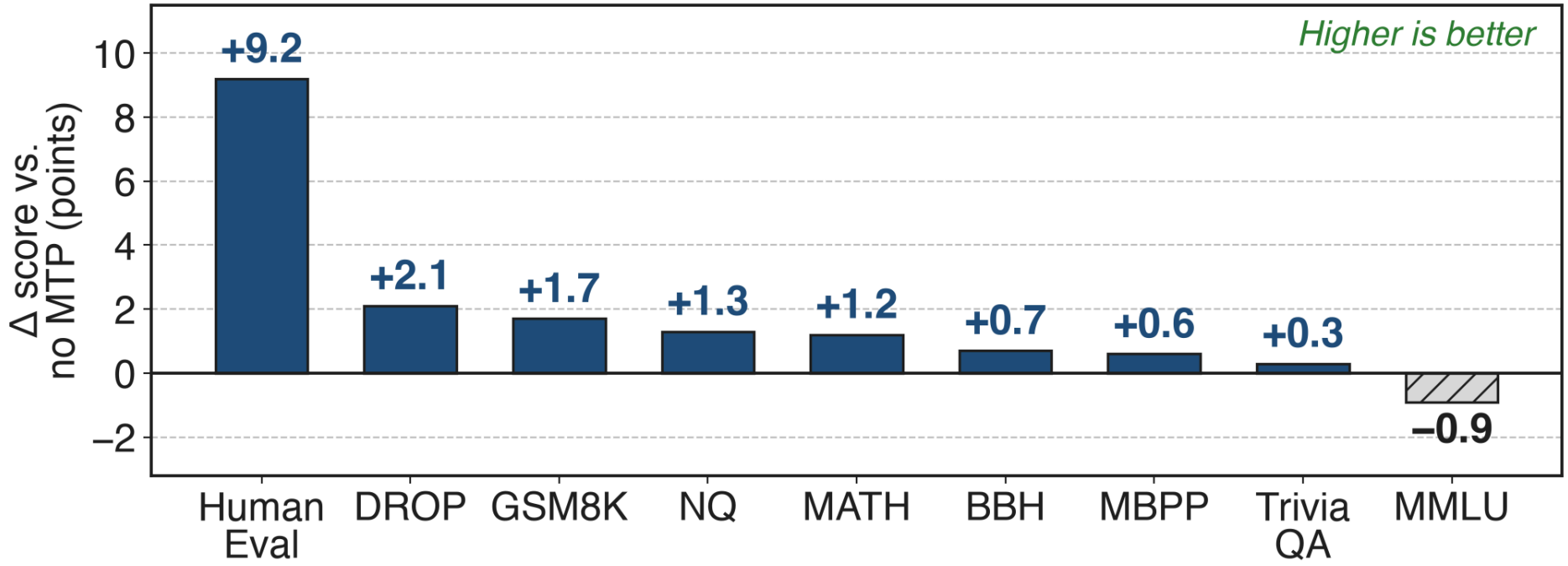
Draft accepted: 2 tokens per pass; rejected: 1, same output either way

❹ MTP is heavy: give it its own stage



MTP outweighs other modules in memory; placement balances stages

MTP adds +9.2 HumanEval at equal inference cost

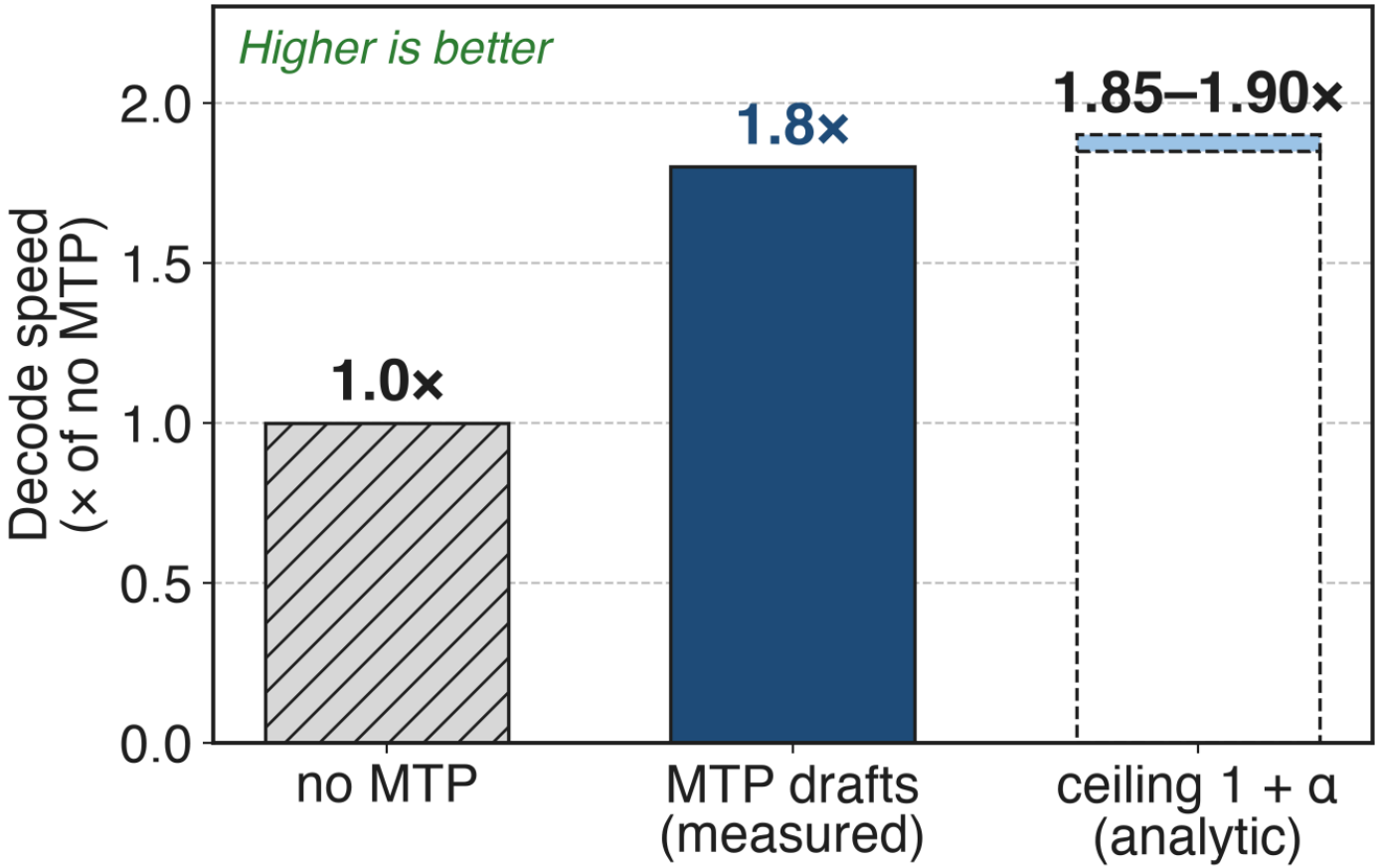


DeepSeek-V3 ablation: 228.7B total / 20.9B active MoE on 540B tokens; MTP discarded at inference

8 of 9 benchmarks improve; MMLU drops 0.9

85–90% acceptance gives 1.8× decode speed

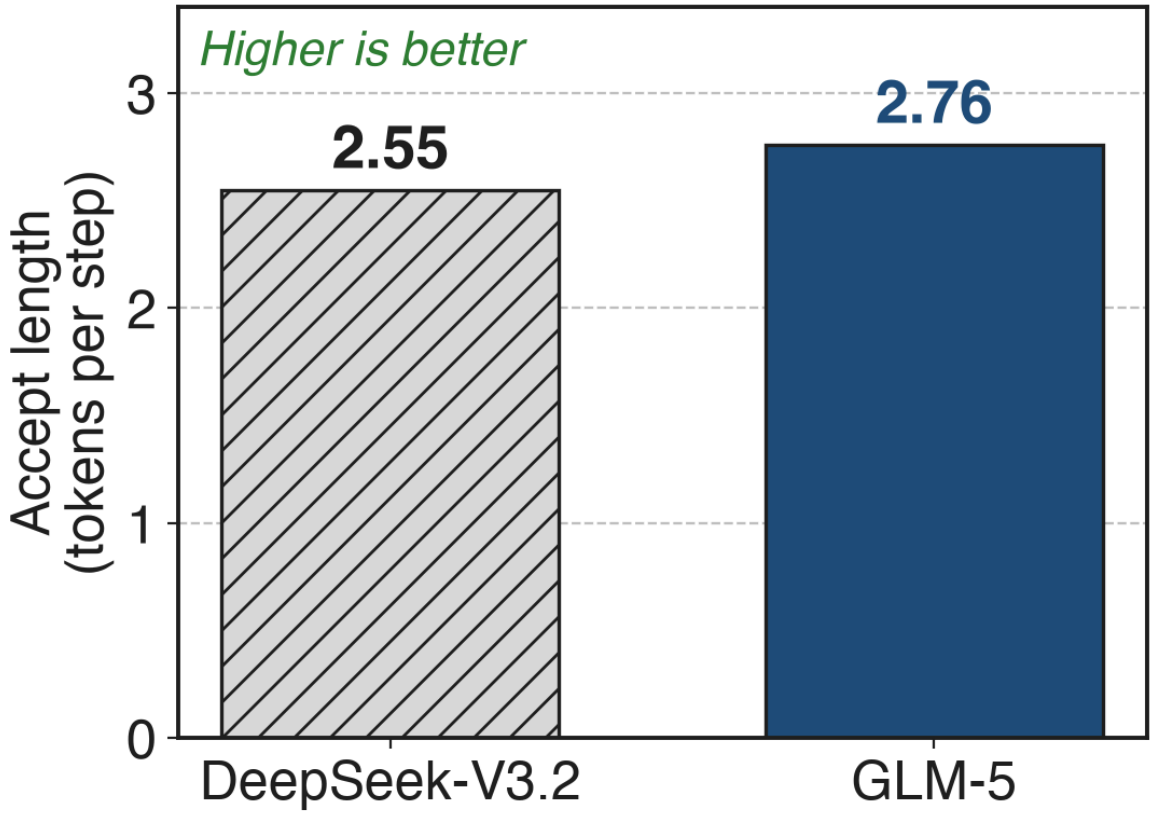
- DeepSeek-V3, one MTP draft per pass
- Acceptance **85–90%**, stable across topics
- Ceiling 1 + α assumes free drafting



Ceiling from Leviathan et al. Eq. 1 / Thm. 3.8 with one draft token (analytic)

Shared 3-step MTP drafts longer: 2.76 vs 2.55

- Trained **1** deep, drafted deeper: later drafts fail more
- GLM-5 shares **3** MTP layers' weights in training
- **4** steps; cross-model, private prompts



The cost: 2% params, extra compute, draft depth

- **+14B params** in DeepSeek-V3 (~**2%**)
- **Extra block + vocab head** every training step
- **≤ 1 MTP layer** with MCore EP overlap
- Deeper drafts need more: shared weights or EAGLE-3 tuning

Summary

- Decode: **1** token per pass; training: **1** target per position
- **Insight**: one block on h + next token both trains and drafts
- **+9.2** HumanEval at equal inference cost (228.7B)
- **85–90%** acceptance → **1.8×** decode speed

https://arxiv.org/abs/2508.06471